

WATER BUDGET DURING REPRODUCTION IN FEMALE COMMON SPINY MICE (*ACOMYS CAHIRINUS*)

A. ALLAN DEGEN,* IRINA S. KHOKHLOVA, MICHAEL KAM, AND IRIS SNIDER

*Desert Animal Adaptations and Husbandry, Wyler Department for Dryland Agriculture,
Jacob Blaustein Institute for Desert Research, Ben-Gurion University of the Negev, Beer-Sheva 84105, Israel*

We determined water budget in the reproducing female common spiny mouse (*Acomys cahirinus*) offered ad libitum millet seeds and water. We hypothesized that the proportional increase in water input in lactating mice compared with nonreproducing mice would be greater than the proportional increase in energy intake, given that more water would be required for higher heat production and for milk production. Females gave birth to 1 or 2 offspring; results of water budget and energy intake did not differ among litter sizes and were combined. Lactating females increased water input and energy intake by 59.1% and 35.2%, respectively, compared with nonreproducing females, confirming our hypothesis. Milk production (1.04 ml/day) accounted for 29% of the difference in water output between lactating and nonreproducing spiny mice.

Key words: *Acomys cahirinus*, evaporative water loss, lactation, milk yield, pregnancy, reproduction, water balance

Water plays a major role in reproduction of rodents. Reproduction is often linked to seasonal rainfall; positive correlations between seasonal rainfall and breeding have been reported in African (Field 1975; Perrin 1980; Perrin and Swanepoel 1987), Australian (Breed 1990), Asian (Rogovin 1985), and North American desert rodents (Kenagy and Bartholomew 1985; Randall 1993). Moreover, populations of free-ranging rabbits decrease in size in arid areas of Australia during droughts when no free water is available and water content of the vegetation declines below about 60% (Richards 1979). Lactation of these rabbits was negatively affected, as were all other aspects of their reproduction. However, as pointed out by several authors (Degen 1997; Perrin and Swanepoel 1987), this does not necessarily mean that rainfall directly initiates reproductive activity, but that the activity may be a response to rainfall and any resultant increase in energy and nutrient supply.

The importance of availability of water alone was demonstrated in reproduction of several desert rodents. Christian (1979) provided permanent sources of drinking in the Namib Desert and monitored the reproductive parameters of several rodent species. Female *Gerbillurus paeba* and *Rhabdomys pumilio* with supplementary water had more pregnancies and a higher proportion of lactating individuals than females with no supplemental water. Furthermore, females with additional

water extended their breeding season further into the hot, dry portion of the year than did controls. The effect of increased availability of water was most pronounced during periods of the year when water stress was greatest. Similarly, reproduction in *Peromyscus truei* increased when artificial sources of water were provided (Bradford 1975). The importance of water to reproduction also is evident in pygmy jerboas, *Salpingotus crassicauda* and *Cardiocranius paradoxus*, in the Gobi desert, which stop breeding during droughts (Rogovin 1985).

Lactating spiny mice (*Acomys cahirinus*) increased their energy intake by 35.2% over nonreproducing mice (Degen et al. 2002). We hypothesized that the proportional increase of water input in these lactating females would be greater than that of energy intake because more water would be required not only for the higher heat production but also for milk production. To test this hypothesis, we determined water budgets of these nonreproducing, pregnant, lactating, and postlactating common spiny mice when offered ad libitum millet seeds and water. We also estimated milk yield.

MATERIALS AND METHODS

Study species and site of capture.—The common spiny mouse (*Acomys cahirinus*; Muridae; average adult body mass of males and females, 35–45 g) is widely distributed among rocky habitats from eastern and northern Africa to Iran (Harrison and Bates 1991; Mendelssohn and Yom-Tov 1987). It is common in Israeli deserts where it inhabits natural crevices among rocks; it does not dig burrows. This omnivorous rodent has a varied diet consisting of moist and dry vegetation, seeds, and animal material (Degen et al. 1986; Happold 1984). In captivity, they must be supplied with water when consuming only seeds (Daily and Haines 1981; Degen and Kam 1992; Shkolnik and Borut 1969).

* Correspondent: degen@bgumail.bgu.ac.il

Reproduction occurs year-round but mainly from the end of winter through late summer (March–September). Gestation is 37–40 days and lactation lasts about 21 days. Females can reproduce 2–3 times per year and litter size ranges between 1 and 4 offspring. Offspring are precocial and are able to move around by their 2nd day (Mendelssohn and Yom-Tov 1987).

Spiny mice were trapped using Sherman traps 6 months prior to the study in the Negev highlands near Sede Boqer (30°52'N, 34°47'E; 475 m altitude). The region is considered an arid zone with mild winters and warm summers (250–300 dry days/year; precipitation: evaporation ratio < 0.2; United Nations Educational, Scientific and Cultural Organization 1977). Mean annual precipitation is 91 mm, all occurring in winter, with large variation among years.

Animal maintenance and measurements.—Our study was conducted at $25 \pm 0.01^\circ\text{C}$ with a photoperiod of 12L:12D. Eighteen adult female spiny mice were offered millet seeds and drinking water ad libitum throughout the study. They were maintained in individual metabolic cages ($11 \times 13 \times 20$ cm) with a wire bottom that allowed for measurement of water and food intake and urine and feces output. Urine was collected under paraffin oil to minimize evaporation. During lactation, the females and offspring were maintained in the metabolic cages for the first 7 days and in larger cages ($13 \times 24 \times 34$ cm) for the next 14 days. The study was approved by the Ben-Gurion University Committee for the Ethical Care and Use of Animals in Experiments.

The study was divided into 4 periods, during which time water input and output and metabolizable energy intake were determined: before pregnancy for 14 days, during pregnancy for 21 days starting 7–10 days after the onset of pregnancy, during the first 7 days of a 21-day lactation (fecal and urine output of the offspring were negligible the first 7 days—Degen et al. 2002), and postweaning for 7 days. Gestation was taken as the 40 days prior to parturition. Details on simultaneous measurement of energy intake on these same females have been presented previously (Degen et al. 2002); included in that report is the growth rate of the offspring.

Following measurements before pregnancy, females were kept together with a number of males for 5 days and then placed in their metabolic cages. If pregnant, they were maintained there until the end of the weaning period. If not pregnant, they were placed with males again. Of the 18 females, 15 became pregnant.

Water input was calculated as preformed water and metabolic water from the seeds plus water drunk. Drinking water was available from a sipper attached to a bottle that could be reached only by the adults. To determine preformed water, daily samples of seeds were weighed and oven-dried at 100°C to constant mass. Metabolic water was calculated from the composition of the seeds and by assuming that 1 g each of metabolized protein, fat, and carbohydrate yielded 0.39, 1.07, and 0.56 g water, respectively (Schmidt-Nielsen 1979). Protein, fat, and carbohydrate of the seeds as a percentage of dry matter were 12.5%, 2.0%, and 78.6%, respectively, according to the supplier (Popli, Kibbutz Saad, Israel). It was assumed that all components were equally digested.

Water output was calculated as water input plus water lost or gained due to changes in body mass. In nonreproducing mice before pregnancy, there was no change in body mass, thus water input equaled water output. Pregnant mice gained body mass; therefore, water output was lower than water input. The retained water was mainly for fetal growth and fetal fluids. A rat fetus has a body water content of close to 90% (Lederman and Rosso 1981) and uterine fluids are >90% water. We assumed that 90% of the body mass gain in pregnant mice was water. Lactating mice lost body mass; therefore, water output was higher than water input. Lost body mass consisted of body water and energy reserves, assumed to be all fat. Given that 1 g

of metabolized fat yields 1.07 g water, then 1 g body mass loss is equivalent to about 1 g (or 1 ml) body water. Female spiny mice gained body mass after pups were weaned; their water input was lower than water output. We assumed that body composition of the spiny mice remained constant and that their total body water was 66% of body mass (Degen et al. 1986). Therefore, the water retained was about 66% of the increase in body mass.

Water output consisted of fecal water, urine water, and evaporative water in nonlactating mice and, in lactating mice, also included milk water. Urine water was calculated by drying samples of urine and multiplying the fraction of water in urine by total urine output. Fecal water was calculated by drying fresh anal fecal samples and by using the fraction of water in fresh feces to calculate total water in feces from total dry matter fecal output. In the 3 groups of nonlactating mice, evaporative water loss (EWL) was calculated as $\text{EWL} = (\text{water output}) - (\text{urine water} + \text{fecal water})$. Evaporative water loss in the lactating spiny mice was calculated from the ratio of EWL : metabolizable energy intake calculated in the other 3 groups. A relationship exists between evaporative water loss and metabolizable energy intake (Degen 1997). We assumed that this held true for the lactating mice, given that in addition to metabolizable energy required for maintenance, 3 groups increased energy intake to support production (Degen 1997): pregnant mice were producing fetuses and uterine tissue, postweaning mice were growing and thus adding body tissue, and lactating mice were producing milk. Milk water was calculated by difference.

Females were weighed (to 0.01 g, model 500 PJ, Precisa, Tongwell, Milton Keynes, United Kingdom) daily before food was offered to determine daily change in body mass.

Statistical analysis.—Time series analyses were used to test whether there were trends in water input and output with time in any of the periods (Kendall 1976). An analysis of covariance was used to examine and remove effects of body mass and litter size on water input and output among periods. Values are presented as means $\pm$ SD, and $P < 0.05$ was chosen as the minimal acceptable level of statistical significance.

RESULTS

There was no apparent trend in water input with time within any of the periods. Therefore, the mean value for each mouse in each period was used in subsequent analyses. Dry matter content of the seeds was $93.5 \pm 0.29\%$ /g dry matter. Fecal water content did not differ among periods and averaged $64.8 \pm 2.27\%$.

Four of the females gave birth to 1 offspring and 11 gave birth to 2 offspring. These females maintained body mass pre-pregnancy, gained body mass during pregnancy, lost body mass during lactation, and gained body mass postweaning (Table 1). Water input and water balance of females that gave birth to either singles or to twins did not differ within any period of measurement and were analyzed together. Water input ($F = 23.6$, $d.f. = 3, 54$, $P < 0.01$) and output differed among periods ($F = 26.4$, $d.f. = 3, 54$, $P < 0.01$) as did all components of input and output. Lactating females increased water input by 59.1% above that of nonpregnant females (Table 1; Fig. 1).

Evaporative water loss (ml/day) as a ratio of metabolizable energy intake (kJ/day) was 0.0792 ml/kJ for nonreproducing females before pregnancy, 0.0784 ml/kJ for pregnant females, and 0.0815 ml/kJ for nonreproducing females after offspring were weaned, for an overall mean of 0.0797 ml/kJ. Using this

TABLE 1.—Body mass (m_b), changes in body mass (Δm_b), metabolizable energy intake (MEI), water input, and water output, including evaporative water loss (EWL), in prepregnant, pregnant, lactating, and postlactating common spiny mice (*Acomys cahirinus*). Lactating spiny mice suckled 1 or 2 pups. Values are means $\pm$ SD, $n = 15$ in each group.

Parameter	Prepregnant	Pregnant	Lactating	Postlactating
Initial m_b (g)	49.4 $\pm$ 7.0	49.4 $\pm$ 7.0	56.2 $\pm$ 8.0	46.1 $\pm$ 5.9
Final m_b (g)	49.4 $\pm$ 7.0	67.7 $\pm$ 6.3	46.1 $\pm$ 9.9	49.1 $\pm$ 5.6
Δm_b (g/day)		0.46 $\pm$ 0.06	−0.48 $\pm$ 0.05	0.21 $\pm$ 0.01
Metabolizable energy intake (kJ/day) ^a	48.6 $\pm$ 4.8	54.7 $\pm$ 6.8	65.7 $\pm$ 10.7	52.9 $\pm$ 7.3
Water input (ml/day)	5.26 $\pm$ 0.85	7.27 $\pm$ 1.31	8.37 $\pm$ 1.40	6.28 $\pm$ 1.28
Water drunk (ml/day)	3.47 $\pm$ 0.70	5.21 $\pm$ 0.94	6.04 $\pm$ 0.87	4.34 $\pm$ 0.75
Preformed water (ml/day)	0.32 $\pm$ 0.07	0.37 $\pm$ 0.07	0.42 $\pm$ 0.08	0.35 $\pm$ 0.05
Metabolic water (ml/day)	1.47 $\pm$ 0.38	1.69 $\pm$ 0.41	1.91 $\pm$ 0.43	1.59 $\pm$ 0.35
Water output (ml/day)	5.26 $\pm$ 0.85	6.86 $\pm$ 1.24	8.85 $\pm$ 1.44	6.13 $\pm$ 1.25
Urine water (ml/day)	0.87 $\pm$ 0.25	1.93 $\pm$ 0.75	1.88 $\pm$ 0.69	1.22 $\pm$ 0.55
Fecal water (ml/day)	0.54 $\pm$ 0.02	0.64 $\pm$ 0.02	0.70 $\pm$ 0.03	0.60 $\pm$ 0.02
Evaporative water loss (ml/day)	3.85 $\pm$ 0.61	4.29 $\pm$ 0.73	5.23 $\pm$ 0.79	4.31 $\pm$ 0.68
Milk water (ml/day)			1.04 $\pm$ 0.21	
EWL/MEI (ml/kJ)	0.0792 $\pm$ 0.001	0.0784 $\pm$ 0.001		0.0815 $\pm$ 0.001

^a From Degen et al. (2002).

value, evaporative water loss for lactating females was calculated as 5.23 ml/day and milk production was calculated as 1.04 ml/day (Table 1; Fig. 1).

DISCUSSION

Lactating females increased their water input by 59.1% (this study) and energy intake by 35.2% (Degen et al. 2002) compared with those of nonreproducing females when measurements were made on the same mice at the same time. This proportionally greater rate in water input than energy intake supported our initial hypothesis.

The increase in water input by lactating female rodents compared with nonreproducing females varies greatly among species. For example, red-backed voles (*Clethrionomys gapperi*) increase water input by 71–131%, white-footed mice (*Peromyscus leucopus*) by 58–129%, gerbils (*Gerbillus allenbyi* and *G. pyramidum*) by 52–136% (Oswald et al., 1993), and *Pseudomys australis* by about 100% (Baverstock and Watts, 1975). In contrast, water input of lactating *Notomys cervinus* females was similar to that of nonlactating females. Similarly, water input of lactating *N. alexis* suckling 3–4 offspring was similar to that of nonlactating females; however, with a litter of 5 offspring, water input of lactating females increased, but not

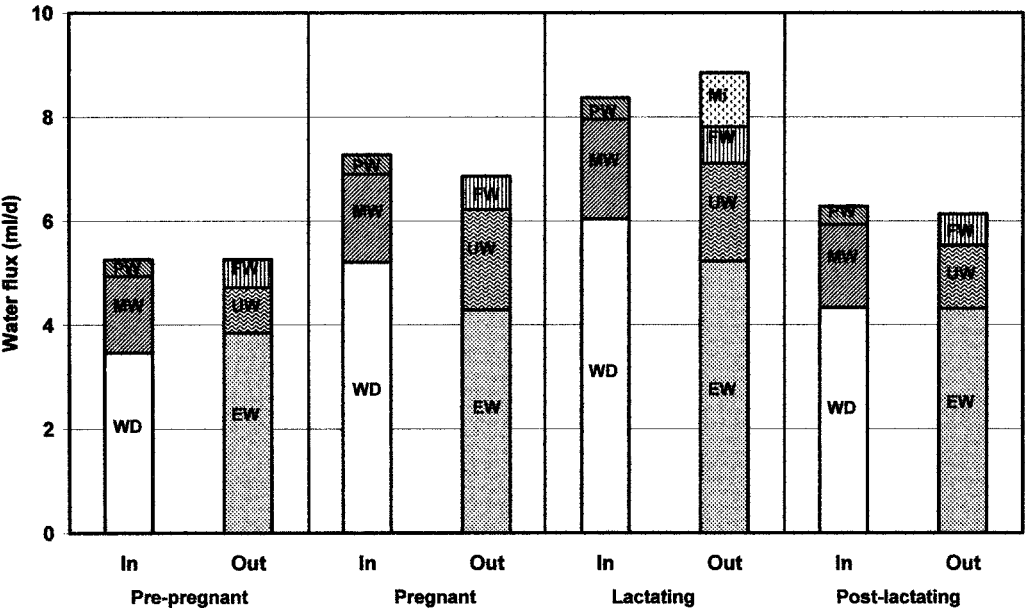


FIG. 1.—Partitioning of the water budget in captive *Acomys cahirinus* females during reproduction. Water input (In) includes preformed (PW) and metabolic (MW) water from food and water drunk (WD); water output (Out) includes urinary (UW), fecal (FW) and evaporative (EW) water loss, and milk (Mi) water. When water input > water output, mice gained body mass; when water output > water input, mice lost body mass.

until 10 days after giving birth. Moreover, water input of lactating *N. mitchelli* females nursing 4 offspring was only slightly higher, but not significantly, than that of nonlactating females (Baverstock and Watts 1975).

Evaporative water loss as a ratio of metabolizable energy intake (ml/kJ) was similar for nonreproducing, pregnant, and postweaning mice. This was not surprising, given that evaporative water loss is related to heat production and heat production is related to metabolizable energy intake (Degen 1997). The mean value in this study of 0.079 ml/kJ was higher than the mean of 0.058 ml/kJ found previously for nonreproducing spiny mice (Kam and Degen 1993). These differences could have been due to dietary differences and water availability. In the latter study, mice received a mixed diet, including snails (which they eat in the wild), and no drinking water, whereas in the present study mice received only seeds and water was available. Evaporative water loss accounted for most of the water output in the spiny mice as it does, in general, in rodents. It accounts for 74–79% of the total water turnover of lactating white-footed mice and for 43–49% of the total water turnover of red-backed voles (Oswald et al. 1993). Evaporative water losses of pregnant and lactating spiny mice, as a proportion of water input, were similar (62.5% and 59.1%, respectively) but lower than evaporative water losses of nonreproducing spiny mice (73.2% prepregnancy and 70.3% postweaning).

Milk yield was estimated as 1.04 ml/day. The proportional increase in water input above that of energy intake was accounted for by the milk volume. Water was available *ad libitum*; thus, spiny mice were easily able to fulfill water needs for lactation. In the field, spiny mice do not drink and receive their water solely from their dietary intake. Presumably, this would entail consuming more snails, which provide much preformed water (Degen et al. 1986; Kam and Degen 1991) and are available year round.

Lactating spiny mice increased energy intake by 35.2% (Degen et al. 2002), a value lower than that found for most lactating rodents, which usually ranges between 65% and 200% above values for nonpregnant females (Gittleman and Thompson 1988; Glazier 1985; Kaczmarski 1966; Mattingly and McClure 1982; Migula 1969; Millar 1978; Poppitt et al. 1994; Randolph et al. 1977). The levels of increase in energy intake by lactating spiny mice was insufficient to maintain body mass and apparently forced them to mobilize energy reserves. The proportional increase in water input in lactating spiny mice was greater than the proportional increase in energy intake. However, if milk production was subtracted from the water input, then the proportional increases in energy and water intakes were similar.

ACKNOWLEDGMENTS

We thank A. Abou-Rachbah for taking care of the mice, R. W. Benjamin for discussions, and 3 anonymous reviewers for helpful suggestions.

LITERATURE CITED

BAVERSTOCK, P. R., AND WATTS, C. H. S. 1975. Water-balance of small lactating rodents. I. *Ad libitum* water intakes and effects of water

- restriction on growth of young. *Comparative Biochemistry and Physiology A. Comparative Physiology* 50:819–825.
- BRADFORD, D. F. 1975. The effects of an artificial water supply on free-living *Peromyscus truei*. *Journal of Mammalogy* 56:705–707.
- BREED, W. G. 1990. Comparative studies on the timing of reproduction and foetal number in six species of Australian rodents (Muridae: Hydromyinae). *Journal of Zoology (London)* 221:1–10.
- CHRISTIAN, D. P. 1979. Comparative demography of three Namib Desert rodents: responses to the provision of supplementary water. *Journal of Mammalogy* 60:679–690.
- DAILY, C. S., AND H. B. HAINES. 1981. Evaporative water loss and water turnover in chronically and acutely water-restricted spiny mice (*Acomys cahirinus*). *Comparative Biochemistry and Physiology A. Comparative Physiology* 68:349–354.
- DEGEN, A. A. 1997. *Ecophysiology of small desert mammals*. Springer-Verlag, Berlin, Germany.
- DEGEN, A. A., AND M. KAM. 1992. Water intake in two coexisting desert rodents, *Acomys cahirinus* and *Gerbillus dasyurus*. *Journal of Mammalogy* 73:201–206.
- DEGEN, A. A., M. KAM, A. HAZAN, AND K. A. NAGY. 1986. Energy expenditure and water flux in three sympatric desert rodents. *Journal of Animal Ecology* 55:421–429.
- DEGEN, A. A., I. S. KHOKHLOVA, M. KAM, AND I. SNIDER. 2002. Energy requirements during reproduction in female common spiny mice (*Acomys cahirinus*). *Journal of Mammalogy* 83:645–651.
- FIELD, A. C. 1975. Seasonal changes in reproduction, diet and body composition in two equatorial rodents. *East African Wildlife Journal* 13:221–235.
- GITTLEMAN, J. L., AND S. D. THOMPSON. 1988. Energy allocation in mammalian reproduction. *American Zoologist* 28:863–875.
- GLAZIER, D. S. 1985. Energetics of litter size in five species of *Peromyscus* with generalizations for other mammals. *Journal of Mammalogy* 66:629–642.
- HAPPOLD, D. C. D. 1984. Small mammals. Pp. 251–257 in *Sahara Desert: key environments* (J. L. Cloudsley-Thompson, ed.). Pergamon Press, London, United Kingdom.
- HARRISON, D. L., AND P. J. BATES. 1991. *The mammals of Arabia*. 2nd ed. Harrison Zoological Museum Publications, Kent, United Kingdom.
- KACZMARSKI, F. 1966. Bioenergetics of pregnancy and lactation in the bank vole. *Acta Theriologica* 11:409–417.
- KAM, M., AND A. A. DEGEN. 1991. Diet selection and energy and water budgets of the common spiny mouse *Acomys cahirinus*. *Journal of Zoology (London)* 225:285–292.
- KAM, M., AND A. A. DEGEN. 1993. Effect of dietary preformed water on energy and water budgets of two sympatric desert rodents, *Acomys russatus* and *Acomys cahirinus*. *Journal of Zoology (London)* 231:51–59.
- KENAGY, G. J., AND G. A. BARTHOLOMEW. 1985. Seasonal reproductive patterns in five coexisting California desert species. *Ecological Monographs* 55:371–397.
- KENDALL, M. G. 1976. Trend. Pp. 39–46 in *Times series*. Hafner, New York.
- LEDERMAN, S. A., AND P. ROSSO. 1981. Effects of fasting during pregnancy on maternal and fetal weight and body composition in well-nourished and undernourished rats. *Journal of Nutrition* 111:1823–1832.
- MATTINGLY, K. K., AND P. A. MCCLURE. 1982. Energetics of reproduction in large-littered cotton rats (*Sigmodon hispidus*). *Ecology* 63:183–195.
- MENDELSSOHN, H., AND Y. YOM-TOV. 1987. *Plants and animals of the land of Israel*, Volume 7, Mammals. Pp. 112–114. Ministry of Defense, the Publishing House, Tel Aviv, Israel (in Hebrew).

- MIGULA, P. 1969. Bioenergetics of pregnancy and lactation in European common vole. *Acta Theriologica* 14:167–179.
- MILLAR, J. S. 1978. Energetics of reproduction in *Peromyscus leucopus*: the cost of lactation. *Ecology* 59:1055–1061.
- OSWALD, C., P. FONKEN, D. ATKINSON, AND M. PALLADINO. 1993. Lactational water balance and recycling in white-footed mice, red-backed voles, and gerbils. *Journal of Mammalogy* 74:963–970.
- PERRIN, M. R. 1980. Ecological strategies of two coexisting rodents. *South African Journal of Science* 76:487–491.
- PERRIN, M. R., AND P. SWANEPOEL. 1987. Breeding biology of the bushveld gerbil *Tatera leucogaster* in relation to diet, rainfall and life history. *South African Journal of Zoology* 22:218–227.
- POPPITT, S. D., J. R. SPEAKMAN, AND P. A. RACEY. 1994. Energetics of reproduction in the lesser hedgehog tenrec, *Echinops telfairi* (Martin). *Physiological Zoology* 67:976–994.
- RANDALL, J. A. 1993. Behavioural adaptations of desert rodents (Heteromyidae). *Animal Behaviour* 45:263–287.
- RANDOLPH, P. A., J. C. RANDOLPH, K. MATTINGLY, AND M. M. FOSTER. 1977. Energy costs of reproduction in the cotton rat, *Sigmodon hispidus*. *Ecology* 58:31–45.
- RICHARDS, G. C. 1979. Variation in water turnover by wild rabbits, *Oryctolagus cuniculus*, in an arid environment, due to season, age group and reproductive condition. *Australian Wildlife Research* 6:289–296.
- ROGOVIN, K. A. 1985. A comparative analysis of behavior in supergeneric groups of jerboas (Rodentia, Dipodidae). *Zoologicheskii Zhurnal* 64:1702–1711 (in Russian).
- SCHMIDT-NIELSEN, K. 1979. *Animal physiology: adaptation and environment*. 2nd ed. Cambridge University Press, Cambridge, United Kingdom.
- SHKOLNIK, A., AND A. BORUT. 1969. Temperature and water relations in two species of spiny mice (*Acomys*). *Journal of Mammalogy* 50:245–245.
- United Nations Educational, Scientific and Cultural Organization. 1977. Map of the world distribution of arid regions. *Man and the Biosphere*, Technical Note 7, Paris, France.

Submitted 10 November 2003. Accepted 30 January 2004.

Associate Editor was Ronald D. Gettinger.